THE INSIGHT

— When the mesh needs backhaul beyond terrestrial reach, our own store-and-forward payload flies.

— Nobody else launching smallsats has a deployed rural ground network waiting — the mesh IS the user-terminal layer.

— A text message or sensor packet needs COVERAGE, not Starlink bandwidth — a handful of CubeSats can blanket Canada.

— Higher orbit is correct for THIS layer: bigger footprint per satellite = fewer needed.

MISSION PROFILE

— Polar LEO constellation blankets Canada incl. North — messaging/sensor tier.

— Latency: store-and-forward hours — right-sized for alerts, notices, telemetry.

— Each sat serves hundreds of nodes per pass — plenty for text/sensor traffic.

— Pathfinder: 2–6 sats prove the constellation, then scale regional.

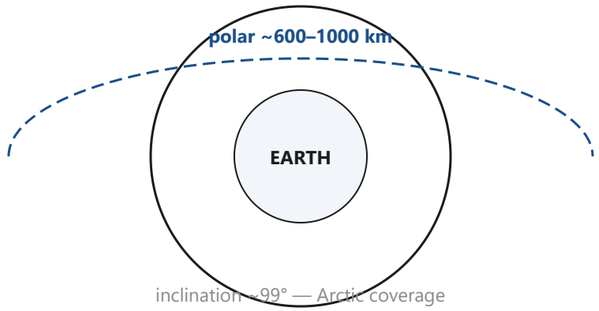
HONEST STATUS

— Vision stage — no satellite exists. The mesh must prove itself first (L1→swarm→revenue).

— This sheet is mission concept, not spacecraft design — payload inherits mesh crypto/routing.

— Every earlier rung funds and justifies this one — the trailhead is the ground network.

ORBIT CONCEPT



inclination ~99° — Arctic coverage

PAYLOAD — DERIVED FROM THE MESH

ELEMENT	IMPLEMENTATION
Radio	UHF/S-band store-and-forward transceiver — LoRa-derived or CC1101-class
Envelope	same signed mesh-net format — satellite is just another hop
Store	onboard flash — holds envelopes until ground station passes
Bus	1U/3U CubeSat (GOMspace/EnduroSat/CubeSat Shop) or smallsat
Ground segment	the deployed mesh + community gateways = user terminals

ASCENT & FUNDING PATH

— Ascent: hosted payload → university CubeSat (CSA) → dedicated smallsat (non-US rideshare) → pathfinder 6–12 sats → regional constellation.

— Funding: CSA CubeSat Initiative · provincial/Indigenous services · IDEaS/defence dual-use · anchor demand for domestic launch.

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